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[p. 335] Also if the discriminant be written

$$K(U) \;=\; \begin{vmatrix} a & h & g & l & j & k \\ h & b & i & f & l & k \\ j & i & c & i & j & l \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \mathfrak{L} & \mathfrak{J} & \mathfrak{K} \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{I} & \mathfrak{F} & \mathfrak{L} & \mathfrak{K} \\ \mathfrak{J} & \mathfrak{I} & \mathfrak{C} & \mathfrak{I} & \mathfrak{J} & \mathfrak{L} \end{vmatrix},$$

then the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$, $\mathfrak{I}$, $\mathfrak{J}$, $\mathfrak{K}$, $\mathfrak{L}$ (equations (20) of § 35) are

$$\begin{aligned}
\mathfrak{A} &= akg + 2fjl - al^2 - gk^2 - fj^2, \\
\mathfrak{B} &= bkh + 2fkl - bl^2 - hf^2 - k^2 i, \\
\mathfrak{C} &= cjf + 2gil - cl^2 - fg^2 - i^2 j, \\
3\mathfrak{F} &= bch + bij - ch^2 + 2gfk - 2bgl + fl^2 - f^2 j - fih, \\
3\mathfrak{G} &= caf + cjk - af^2 + 2hgi - 2chl + gl^2 - g^2 k - gjf, \\
3\mathfrak{H} &= abg + aki - bg^2 + 2fhj - 2afl + hl^2 - h^2 i - hkg, \\
3\mathfrak{I} &= bcj + cfh - bg^2 + 2kig - 2ckl + jl^2 - j^2 h - fij, \\
3\mathfrak{J} &= cak + agf - ch^2 + 2ijh - 2ail + kl^2 - k^2 f - gjk, \\
3\mathfrak{K} &= abi + bhg - ai^2 + 2jkf - 2bjl + il^2 - i^2 g - hki, \\
6\mathfrak{L} &= aic + 3fgh + 3ijk + 2l^3 - afi - bgj - chk - 2lgk - 2lhi - 2lfj.
\end{aligned}$$

[The equation $K(U) = R = 64\,S^3 - T^2$ would however afford a perhaps easier formula for the calculation of the discriminant.]

145. A Memoir upon Caustics.

[*From the Philosophical Transactions of the Royal Society of London*, vol. CXLVII. for the year 1857, pp. 273–312. Received May 1,—Read May 8, 1856.]

[p. 336] The following memoir contains little or nothing that can be considered new in principle; the object of it is to collect together the principal results relating to caustics *in plano*, the reflecting or refracting curve being a right line or a circle, and to discuss, with more care than appears to have been hitherto bestowed upon the subject, some of the more remarkable cases. The memoir contains in particular researches relating to the caustic by refraction of a circle for parallel rays, the caustic by reflexion of a circle for rays proceeding from a point, and the caustic by refraction of a circle for rays proceeding from a point; the result in the last case is not worked out, but it is shown how the equation in rectangular coordinates is to be obtained by equating to zero the discriminant of a rational and integral function of the sixth degree. The memoir treats also of the secondary caustic, or orthogonal trajectory of the reflected or refracted rays, in the general case of a reflecting or refracting circle and rays proceeding from a point; the curve in question, or rather a secondary caustic, is, as is well known, the Oval of Descartes or 'Cartesian': the equation is discussed by a method which gives rise to some forms of the curve which appear to have escaped the notice of geometers. By considering the caustic as the evolute of the secondary caustic, it is shown that the caustic, in the general case of a reflecting or refracting circle and rays proceeding from a point, is a curve of the sixth class only. The concluding part of the memoir treats of the curve which, when the incident rays are parallel, must be taken for the secondary caustic in the place of the Cartesian, which, for the particular case in question, passes off to infinity. In the course of the memoir, I reproduce a theorem first given, I believe, by me in the *Philosophical Magazine*, viz. that there are six different systems of a radiant point and refracting circle which give rise to identically the same caustic, [see post, XXVIII]. **[p. 337]** The memoir is divided into sections, each of which is to a considerable extent intelligible by itself, and the subject of each section is for the most part explained by the introductory paragraph or paragraphs.

I.

Consider a ray of light reflected or refracted at a curve, and suppose that ξ, η are the coordinates of a point Q on the incident ray, α, β the coordinates of the point G of incidence upon the reflecting or refracting curve, a, b the coordinates of a point N upon the normal at the point of incidence, x, y the coordinates of a point q on the reflected or refracted ray.

Write for shortness,

$$(b-\beta)(\xi-\alpha)-(a-\alpha)(\eta-\beta)=\nabla\, QGN,$$
$$(a-\alpha)(\xi-\alpha)+(b-\beta)(\eta-\beta)=\square\, QGN,$$

$\nabla\, QGN$ then is equal to twice the area of the triangle QGN, and if ξ, η instead of being the coordinates of a point Q on the incident ray were current coordinates, the equation $\nabla\, QGN=0$ would be the equation of the line through the points G and N, i.e. of the normal at the point of incidence; and in like manner the equation $\square\, QGN=0$ would be the equation of the line through G perpendicular to the line through the points G and N, i.e. of the tangent at the point of incidence.

We have

$$\overline{NG}^2=(a-\alpha)^2+(b-\beta)^2,$$

and therefore identically,

$$\overline{NG}^2\cdot\overline{QG}^2=\nabla\, QGN^2+\square\, QGN^2.$$

Suppose for a moment that ϕ is the angle of incidence and ϕ' the angle of reflexion or refraction; and let μ be the index of refraction (in the case of reflexion $\mu=-1$), then writing

$$(b-\beta)(x-\alpha)-(a-\alpha)(y-\beta)=\nabla\, qGN,$$
$$(a-\alpha)(x-\alpha)+(b-\beta)(y-\beta)=\square\, qGN,$$

and

$$\overline{qG}^2=(x-\alpha)^2+(y-\beta)^2,$$

we have

$$\sin\phi = \frac{\nabla\, QGN}{\overline{NG}\cdot\overline{GQ}}, \qquad \sin\phi' = \frac{\nabla\, qGN}{\overline{NG}\cdot\overline{Gq}},$$

and substituting these values in the equation

$$\sin^2\phi - \mu^2\sin^2\phi' = 0,$$

[p. 338] we obtain

$$\overline{qG}^2\,\nabla\, QGN^2 - \mu^2\,\overline{QG}^2\,\nabla\, qGN^2 = 0,$$

an equation which is rational of the second order in x, y, the coordinates of a point q on the refracted ray; this equation must therefore contain, as a factor, the equation of the refracted ray; the other factor gives the equation of a line equally inclined to, but on the opposite side of the normal; this line (which of course has no physical existence) may be termed the *false refracted ray.* The caustic is geometrically the envelope of the pair of rays, and for finding the equation of the caustic it is obviously convenient to take the equation of the two rays conjointly in the form under which such equation has just been found, without attempting to break the equation up into its linear factors.

It is however interesting to see how the resolution of the equation may be effected; for this purpose multiply the equation by $\overline{NG}^2$, then reducing by means of a previous formula, the equation becomes

$$\big(\nabla\, qGN^2 + \Box\, qGN^2\big)\nabla\, QGN^2 - \mu^2\big(\nabla\, QGN^2 + \Box\, QGN^2\big)\nabla\, qGN^2 = 0,$$

which is equivalent to

$$\nabla\, qGN\big(\mu^2\,\Box\, QGN + (\mu^2-1)\nabla\, QGN\big) - \Box\, qGN\cdot\nabla\, QGN = 0,$$

and the factors are

$$\nabla\, qGN\sqrt{\mu^2\,\Box\, QGN^2 + (\mu^2-1)\,\nabla\, QGN^2} \pm \Box\, qGN\cdot\nabla\, QGN = 0;$$

it is in fact easy to see that these equations represent lines passing through the point G and inclined to GN at angles $\pm\phi'$, where ϕ' is given by the equations

$$\sin\phi = \mu\sin\phi', \qquad \tan\phi' = \frac{\nabla\, QGN}{\Box\, QGN},$$

and there is no difficulty in distinguishing in any particular case between the refracted ray and the false refracted ray.

In the case of reflexion $\mu = -1$, and the equations become

$$\nabla\, qGN\cdot\Box\, QGN \pm \Box\, qGN\cdot\nabla\, QGN = 0;$$

the equation

$$\nabla\, qGN\cdot\Box\, QGN - \Box\, qGN\cdot\nabla\, QGN = 0$$

is obviously that of the incident ray, which is what the false refracted ray becomes in the case of reflexion; and the equation

$$\nabla\, qGN\cdot\Box\, QGN + \Box\, qGN\cdot\nabla\, QGN = 0$$

is that of the reflected ray.

[p. 339]

II.

But instead of investigating the nature of the caustic itself, we may begin by finding the secondary caustic or orthogonal trajectory of the refracted rays, i.e. a curve having the caustic for its evolute; suppose that the incident rays are all of them normal to a certain curve, and let Q be a point upon this curve, and considering the ray through the point Q, let G be the point of incidence upon the refracting curve; then if the point G be made the centre of a circle the radius of which is $\mu^{-1}\cdot GQ$, the envelope of the circles will be the secondary caustic. It should be noticed that, if the incident rays proceed from a point, the most simple course is to take such point for the point Q. The remark, however, does not apply to the case where the incident rays are parallel; the point Q must here be considered as the point in which the incident ray is intersected by some line at right angles to the rays, and there is not in general any one line which can be selected in preference to another. But if the refracting curve be a circle, then the line perpendicular to the incident rays may be taken to be a diameter of the circle. To translate the construction into analysis, let ξ, η be the coordinates of the point Q, and α, β the coordinates of the point G, then ξ, η, α, β are in effect functions of a single arbitrary parameter; and if we write

$$\overline{GQ}^2 = (\xi - \alpha)^2 + (\eta - \beta)^2, \qquad \overline{Gq}^2 = (x - \alpha)^2 + (y - \beta)^2,$$

then the equation

$$\mu^2\,\overline{Gq}^2 - \overline{GQ}^2 = 0,$$

where x, y are to be considered as current coordinates, and which involves of course the arbitrary parameter, is the equation of the circle, and the envelope is obtained in the usual manner. This is the well-known theory of Gergonne and Quetelet.

III.

There is however a simpler construction of the secondary caustic in the case of the reflexion of rays proceeding from a point. Suppose, as before, that Q is the radiant point, and let G be the point of incidence. On the tangent at G to the reflecting curve, let fall a perpendicular from Q, and produce it to an equal distance on the other side of the tangent; then if q be the extremity of the line so produced, it is clear that q is a point on the reflected ray Gq, and it is easy to see that the locus of q is the secondary caustic. Produce now QG to a point Q' such that $GQ' = QG$, it is clear that the locus of Q' will be a curve similar to and similarly situated with and twice the magnitude of the reflecting curve, and that the two curves have the point Q for a centre of similitude. And the tangent at Q' passes through the point q, i.e. q is the foot of the perpendicular let fall from Q upon the tangent at Q'; we have therefore the theorem due to Dandelin, viz.

[p. 340] If rays proceeding from a point Q are reflected at a curve, then the secondary caustic is the locus of the feet of the perpendiculars let fall from the point Q upon the tangents of a curve similar to and similarly situated with and twice the magnitude of the reflecting curve, and such that the two curves have the point Q for a centre of similitude.

IV.

If rays proceeding from a point Q are reflected at a line, the reflected rays will proceed from a point q situate on the perpendicular let fall from Q, and at an equal distance on the other side of the reflecting line. The point q may be spoken of as the image of Q; it is clear that if Q be considered as a variable point, then the locus of the image q will be a curve equal and similar but oppositely situated to the curve, the locus of Q, and which may be spoken of as the image of such curve. Hence it at once follows, that if the incidental rays are tangent, or normal, or indeed in any other manner related to a curve, then the reflected rays will be tangent, or normal, or related in a corresponding manner to a curve the image of the first-mentioned curve. The theory of the combined reflexions and refractions of a pencil of rays transmitted through a plate or prism, is, by the property in question, rendered very simple. Suppose, for instance, that a pencil of rays is refracted at the first surface of a plate or prism, and after undergoing any number of internal reflexions, finally emerges after a second refraction at the first or second surface; in order to find the caustic enveloped by the rays after the second refraction, it is only necessary to form the successive images of the first caustic corresponding to the different reflexions, and finally to determine the caustic for refraction in the case where the incident rays are the tangents of the caustic which is the last of the series of images; the problem is not in effect different from that of finding the caustic for refraction in the case where the incident rays are the tangents to the caustic after the first refraction, but the line at which the second refraction takes place is arbitrarily situate with respect to the caustic. Thus e.g. suppose the incident rays proceed from a point, the caustic after the first refraction is, it will be shown in the sequel, the evolute of a conic; for the complete theory of the combined reflexions and refractions of the pencil by a plate or prism, it is only necessary to find the caustic by refraction, where the incident rays are the normals of a conic, and the refracting line is arbitrarily situate with respect to the conic.

V.

Suppose that rays proceeding from a point Q are refracted at a line; and take the refracting line for the axis of y, the axis of x passing through the radiant point Q, and take the distance QA for unity. Suppose that the index of refraction μ is put equal to $\frac{1}{k}$. Then if ϕ be the angle of incidence and ϕ' the angle of refraction, **[p. 341]** we have $\sin\phi' = k\sin\phi$, and the equation $y = x\tan\phi' - \tan\phi$ of the refracted ray becomes, putting for ϕ' its value,

$$y - \frac{k\sin\phi}{\sqrt{1-k^2\sin^2\phi}}\,x - \tan\phi = 0.$$

[Figure: incident ray from Q on the axis of x refracted at the axis of y; angles of incidence and refraction at A.]

Differentiating with respect to the variable parameter and combining the two equations, we obtain, after a simple reduction,

$$kx = -\frac{(1-k^2\sin^2\phi)^{\frac{3}{2}}}{\cos^3\phi}, \qquad k'y = -\frac{k'^2\sin^3\phi}{\cos^3\phi},$$

where $k' = \sqrt{1-k^2}$; hence eliminating,

$$(kx)^{\frac{2}{3}} - (k'y)^{\frac{2}{3}} = 1,$$

which is the equation of the caustic. When the refraction takes place into a denser medium k is less than 1, and k'^2 is positive, the caustic is therefore the evolute of a hyperbola (see fig. 1); but when the refraction takes place in a rarer medium k is greater than 1, and k'^2 is negative, the caustic is therefore the evolute of an ellipse (see fig. 2). These results appear to have been first obtained by Gergonne. The conic (hyperbola or ellipse) is the secondary caustic, and as such may be obtained as follows.

VI.

The equation of the variable circle is

$$x^2 + (y-\tan\phi)^2 - k^2\sec^2\phi = 0;$$

or reducing, the equation is

$$x^2 + y^2 - 2y\tan\phi - k^2\tan^2\phi + 1 - k^2 = 0;$$

whence, considering $\tan\phi$ as the variable parameter, the equation of the envelope is

$$k^2(x^2+y^2-k^2) - y^2 = 0,$$

that is,

$$k^2y^2 - k'^2x^2 - k^2k'^2 = 0,$$

[p. 342] or

$$\frac{x^2}{k^2} - \frac{y^2}{k'^2} = 1$$

is the equation of the secondary caustic, or conic having the caustic for its evolute. The radiant point, it is clear, is a focus of the conic.

VII.

Let the equation of the refracted ray be represented by

$$Xx + Yy + Z = 0,$$

we have

$$X : Y : Z = \frac{-k \sin\phi}{\sqrt{1 - k^2 \sin^2\phi}} \;:\; 1 \;:\; -\tan\phi,$$

from which we obtain

$$\frac{k^2}{X^2} - \frac{k'^2}{Y^2} = \frac{1}{Z^2}$$

for the tangential equation of the caustic; or if we represent the equation of the refracted ray by

$$Xx + Yy - k = 0,$$

then we have

$$\frac{k^2}{X^2} - \frac{k'^2}{Y^2} = \frac{1}{k^2}$$

for the tangential equation of the caustic.

[Figure 1: Caustic by refraction of a line into a denser medium, evolute of a hyperbola; refracted-ray family tangent to the hyperbolic arc.] *[Figure 2: Caustic by refraction of a line into a rarer medium, evolute of an ellipse; cusped caustic inside a circle.]*

[p. 343]

VIII.

If a ray be reflected at a circle; we may take a, b as the coordinates of the centre of the circle, and supposing as before that ξ, η are the coordinates of a point Q in the incident ray, α, β the coordinates of the point G of incidence, and x, y the coordinates of a point q in the reflected ray, the equation of the reflected ray, treating x, y as current coordinates, is

$$\begin{aligned}&\{(b-\beta)(\xi-\alpha) - (a-\alpha)(\eta-\beta)\}\{(a-\alpha)(\xi-\alpha) + (b-\beta)(\eta-\beta)\}\\ &\quad + \{(a-\alpha)(x-\alpha) + (b-\beta)(y-\beta)\}\{(b-\beta)(\xi-\alpha) - (a-\alpha)(\eta-\beta)\} = 0.\end{aligned}$$

Write for shortness,

$$\begin{aligned}N_{Q,G} &= (b-\beta)(x-\alpha) - (a-\alpha)(y-\beta),\\ T_{Q,G} &= (a-\alpha)(x-\alpha) + (b-\beta)(y-\beta),\end{aligned}$$

and similarly for $N_{q,G}$ &c.; the equation of the reflected ray is

$$N_{q,G} \cdot T_{Q,G} + T_{q,G} \cdot N_{Q,G} = 0.$$

Suppose that the reflected ray meets the circle again in G' and undergoes a second reflexion, and let x', y' be the coordinates of a point q' in the ray thus twice reflected. We see first (G' being a point in the first reflected ray) that

$$N_{q,G'} \cdot T_{Q,G'} + T_{q,G'} \cdot N_{Q,G'} = 0.$$

Again, considering G as a point in the ray by the reflexion of which the second reflected ray arises, the equation of the second reflected ray is

$$N_{q',G'} \cdot T_{G,G'} + T_{q',G'} \cdot N_{G,G'} = 0;$$

and from the form of the expressions $N_{a,b}$, $T_{a,b}$ it is clear that

$$N_{a,b} = -N_{b,a}, \qquad T_{a,b} = T_{b,a};$$

the equation for the second reflected ray may therefore be written under the form

$$N_{q',G'} \cdot T_{Q,G'} - T_{q',G'} \cdot N_{Q,G'} = 0;$$

or reducing by a previous equation, we obtain finally for the equation of the second reflected ray,

$$N_{q',G'} \cdot T_{Q,G} + T_{q',G'} \cdot N_{Q,G} = 0,$$

and in like manner the equation for the third reflected ray is

$$N_{q'',G''} \cdot T_{Q,G} + T_{q'',G''} \cdot N_{Q,G} = 0,$$

and so on, the equation for the last reflected ray containing, it will be observed, the coordinates of the radiant point and of the first and last points of incidence (the coordinates of the last point of incidence can of course only be calculated from those of the radiant point and the first point of incidence, through the coordinates of the intermediate points of incidence), but not containing explicitly the coordinates of any of the intermediate points of incidence. The form is somewhat remarkable, but the result is really the same with that obtained by simple geometrical considerations, as follows.

[p. 344]

IX.

Consider a ray reflected any number of times at a circle; and let G_0G_1 be the ray incident at G_1, and GG' the last reflected ray, the point at which the reflexion takes place or last point of incidence being G. Take the centre O of the circle for the origin, and any two lines Ox, Oy through the centre and at right angles to each other for axes, and let Ox meet the circle in the point A. Write

$$\begin{aligned} \angle AOG_1 &= \theta_0, & \angle xG_0G_1 &= \psi_0, \\ \angle AOG &= \theta, & \angle xGG' &= \psi, \\ \angle G_1G_2O &= \phi; \end{aligned}$$

then the radius of the circle being taken as unity and its centre as origin, the equation of the reflected ray is

$$y - \sin\theta = \tan\psi\,(x - \cos\theta);$$

and if there have been n reflexions, then

$$\theta = \theta_0 + n(\pi - 2\phi) = \theta_0 + n\pi - 2n\phi,$$

$$\psi = \psi_0 - 2n\phi,$$

and therefore the equation of the reflected ray is

$$x\cos(\psi_0 - 2n\phi) - y\sin(\psi_0 - 2n\phi) + (-)^n \sin(\psi_0 - \theta_0) = 0.$$

[Figure: circle centred at O with radius OA along the x-axis; incident ray G_0G_1, intermediate points of incidence G_1, G_2, …, and the final point G from which the last reflected ray GG' emerges.]

X.

If a pencil of parallel rays is reflected any number of times at a circle, then taking AO for the direction of the incident rays, we may write $\theta_0 = \phi$, $\psi_0 = \pi$, and the equation of a reflected ray is

$$x\sin 2n\phi + y\cos 2n\phi = (-)^n \sin\phi;$$

[p. 345] differentiating with respect to the variable parameter, we find

$$x\cos 2n\phi - y\sin 2n\phi = \frac{(-)^n}{2n}\cos\phi;$$

and these equations give

$$x = \frac{(-)^n}{4n}\{(2n+1)\cos(2n-1)\phi - (2n-1)\cos(2n+1)\phi\},$$
$$y = \frac{(-)^n}{4n}\{-(2n+1)\sin(2n-1)\phi + (2n-1)\sin(2n+1)\phi\},$$

which may be taken for the equation of the caustic; the caustic is therefore an epicycloid: this is a well-known result.

XI.

If rays proceeding from a point upon the circumference are reflected any number of times at a circle, then taking the point A for the radiant point, we have $\theta_0 = 0$, $\psi_0 = \pi - \phi$, and the equation of a reflected ray is

$$x\sin(2n+1)\phi + y\cos(2n+1)\phi = (-)^n \sin\phi;$$

differentiating with respect to the variable parameter, we find

$$x\cos(2n+1)\phi - y\sin(2n+1)\phi = \frac{(-)^n}{2n+1}\sin\phi;$$

and these equations give

$$x = \frac{(-)^n}{2n+1}\{(n+1)\cos 2n\phi - n\cos(2n+2)\phi\},$$
$$y = \frac{(-)^n}{2n+1}\{-(n+1)\sin 2n\phi + n\sin(2n+2)\phi\},$$

which may be taken as the equation of the caustic; the caustic is therefore in this case also an epicycloid: this is a well-known result.

XII.

Consider a pencil of parallel rays refracted at a circle; take the radius of the circle as unity, and let the incident rays be parallel to the axis of x, then if ϕ, ϕ' be the angles of incidence and refraction, and μ or $\frac{1}{k}$ be the index of refraction, so that $\sin\phi' = k\sin\phi$, the coordinates of the point of incidence are $\cos\phi$, $\sin\phi$, and the equation of the refracted ray is

$$y - \sin\phi = \tan(\phi - \phi')\,(x - \cos\phi),$$

i.e.

$$\cos(\phi-\phi')(y-\sin\phi) = \sin(\phi-\phi')(x-\cos\phi),$$

[p. 346] or

$$x\sin(\phi-\phi') - y\cos(\phi-\phi') = \sin\phi',$$

which may also be written

$$\sin\phi\,(y\cos\phi - x\sin\phi)\cos\phi' + (y\sin\phi + x\cos\phi - 1)\sin\phi' = 0.$$

Writing $k\sin\phi$, $\sqrt{1-k^2\sin^2\phi}$ instead of $\sin\phi'$, $\cos\phi'$, and putting for shortness

$$y\cos\phi - x\sin\phi = Y, \qquad x\cos\phi + y\sin\phi = X, \qquad \frac{k\sin\phi}{\sqrt{1-k^2\sin^2\phi}} = \Phi,$$

the equation of the refracted ray becomes

$$Y + \Phi(X-1) = 0;$$

and differentiating with respect to the variable parameter ϕ, observing that

$$\frac{dY}{d\phi} = -X, \qquad \frac{dX}{d\phi} = Y, \qquad \frac{d\Phi}{d\phi} = \frac{k\cos\phi}{(1-k^2\sin^2\phi)^{\frac{3}{2}}} = \frac{\cot\phi}{1-k^2\sin^2\phi}\,\Phi,$$

we have

$$-X + \Phi\Big(Y + \frac{\cot\phi\,(X-1)}{1-k^2\sin^2\phi}\Big) = 0,$$

and the combination of the two equations gives

$$Y = \frac{\Phi\,(1-k^2\sin^2\phi)}{\Phi\cot\phi - 1}, \qquad X = \frac{\Phi\cot\phi - k^2\sin^2\phi}{\Phi\cot\phi - 1},$$

and we have therefore

$$y = Y\cos\phi + X\sin\phi = \frac{\Phi\,(1-k^2\sin^2\phi)\cos\phi + \sin\phi\,(\Phi\cot\phi - k^2\sin^2\phi)}{\Phi\cot\phi - 1} = k^2\sin^3\phi,$$

$$x = -Y\sin\phi + X\cos\phi = \frac{-\Phi\,(1-k^2\sin^2\phi)\sin\phi + \cos\phi\,(\Phi\cot\phi - k^2\sin^2\phi)}{\Phi\cot\phi - 1},$$

i.e.

$$x = \frac{\Phi\,(1-k^2\sin^2\phi) - k^2\sin^2\phi\cos\phi}{-\Phi\cos\phi\sin\phi}\;\cos\phi;$$

[p. 347] or multiplying the numerator and denominator by $(1-k^2\sin^2\phi)^{\frac{1}{2}}\,(\Phi\cos\phi+\sin\phi)$, the numerator becomes

$$\begin{aligned}
&-\Phi\,(1-k^2\sin^2\phi)\big\{\Phi^2\cos^2\phi - k^2\sin^2\phi\cos\phi\\
&\qquad + \Phi\,(\sin\phi\,(1-k^2\sin^2\phi) - k^2\sin^3\phi\cos\phi)\big\}\\
&= -k^2\sin^2\phi\cos\phi\,\big\{(1-k^2\sin^2\phi) - \sin^2\phi\,(1-k^2\sin^2\phi)\big\}\\
&\qquad + k\sin^3\phi\sqrt{1-k^2\sin^2\phi}\,(1-k^2\sin^2\phi)\\
&= \Phi\,(1-k^2\sin^2\phi)\cos^3\phi - k\sin^3\phi\,(1-k^2\sin^2\phi)^{\frac{3}{2}},
\end{aligned}$$

and the denominator becomes

$$k^2\sin^2\phi\cos^2\phi - (1-k^2\sin^2\phi)\sin^2\phi = -k'^2\sin^2\phi,$$

if $k'^2 = 1-k^2$.

Hence we have for the coordinates of the point of the caustic,

$$\begin{aligned}
k'^2x &= -k'^2\cos^3\phi - k\,(1-k^2\sin^2\phi)^{\frac{3}{2}},\\
k'^2y &= k^2\sin^3\phi,
\end{aligned}$$

and eliminating ϕ, we obtain for the equation of the caustic,

$$k'^2x = -k'^2\big(1-k^{-\frac{2}{3}}y^{\frac{2}{3}}\big)^{\frac{3}{2}} - k\big(1-k^{\frac{2}{3}}y^{\frac{2}{3}}\big)^{\frac{3}{2}};$$

or writing $\dfrac{1}{\mu}$ instead of k, we find

$$\Big(1-\frac{1}{\mu^2}\Big)x = -\big(1-\mu^{\frac{2}{3}}y^{\frac{2}{3}}\big)^{\frac{3}{2}} + \mu\big(1-\mu^{-\frac{2}{3}}y^{\frac{2}{3}}\big)^{\frac{3}{2}}$$

for the equation of the caustic by refraction of the circle, for parallel rays. The equation was first obtained by St Laurent.

XIII.

The discussion of the preceding equation presents considerable interest. In the first place to obtain the rational form write

$$x = \alpha, \qquad \bigl(1 - \mu^{\frac{2}{3}} y^{\frac{2}{3}}\bigr)^{\frac{1}{2}} = \beta, \qquad \mu\bigl(1 - \mu^{-\frac{2}{3}} y^{\frac{2}{3}}\bigr)^{\frac{1}{2}} = \gamma;$$

this gives

$$\alpha^4 + 2\alpha^2(\beta^2 + \gamma^2) + (\beta^2 - \gamma^2)^2 = 0,$$

and we have

$$\beta^2 + \gamma^2 = 1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + 3\mu^{\frac{4}{3}} y^{\frac{4}{3}} - \mu^2\, y^2,$$
$$\gamma^2 = \mu^2 - 3\mu^{\frac{4}{3}} y^{\frac{2}{3}} + 3\mu^{\frac{2}{3}} y^{\frac{4}{3}} - y^2,$$

and consequently

$$\beta^4 - \gamma^4 = (1 - \mu^2)\bigl[1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + (1 + \mu^2)\, y^{\frac{4}{3}}\bigr].$$

[p. 348] Hence dividing out by the factor $(1 - \mu^2)^2$, the equation becomes

$$(1 - \mu^2)^2\, x^4 + 2\bigl(1 + \mu^2 - 6\mu^2 y^{\frac{2}{3}} + 3\mu^{\frac{2}{3}}(1 + \mu^2)\, y^{\frac{4}{3}} - (1 + \mu^2)\, y^2\bigr)\, x^2 + \bigl(1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + (1 + \mu^2)\, y^{\frac{4}{3}}\bigr)^2 = 0;$$

or reducing and arranging,

$$\begin{aligned}
(1 + \mu^2)^2\, x^4 + 2(1 + \mu^2)\, x^2\, &\bigl[(1 + \mu^2)\, x^2 + (1 + \mu^2)\, y^2\bigr] \\
&+ 2(1 + \mu^2)\, x^2\, y^2 + (1 + \mu^2)^2\, y^4 \\
&+ 2\mu^{\frac{2}{3}}\bigl(6\mu^4 - 9\mu^2\,(1 + \mu^2)\bigr)\, y^{\frac{2}{3}} \\
&+ 2\mu^{\frac{4}{3}}\bigl(6\mu^4 + 6\mu^2\,(1 + \mu^2)\bigr)\, y^{\frac{4}{3}} = 0,
\end{aligned}$$

which is of the form

$$A + 3\mu^{\frac{2}{3}}\, B\, y^{\frac{2}{3}} - 6\mu^2\, C\, y^{\frac{4}{3}} = 0;$$

and the rationalized equation is

$$A^3 + 27\mu^2\, B^3\, y^2 - 216\mu^6\, C^3\, y^4 + 54\mu^4\, ABC\, y^2 = 0,$$

where the values of A, B, C may be written

$$\begin{aligned}
&A = (1 + \mu^2)^2\,(x^2 + y^2)^2 - 2\,(1 + \mu^2)\,(x^2 + y^2) + 1, \\
&B = x^2 + 3y^2, \\
&C = (1 + \mu^2)\,(x^2 + y^2) + 1,
\end{aligned}$$

the caustic is therefore a curve of the 12th order.

To find where the axis of x meets the curve, we have

$$y = 0, \quad A^3 = 0,$$

where

$$A_0 = -(1 + \mu^2)^2\, x^4 + 2\,(1 + \mu^2)\, x^2 - 1,$$

i.e.

$$\bigl[(1 + \mu)^2\, x^2 - 1\bigr]\,\bigl[(1 - \mu)^2\, x^2 - 1\bigr] = 0,$$

or

$$x = \pm\,\frac{1}{1 + \mu}, \qquad x = \pm\,\frac{1}{1 - \mu},$$

or there are in all four points, each of them a point of triple intersection.

To find where the line ∞ meets the curve, we have

$$y = \infty, \quad A'^3 = 0,$$

where

$$A' = (x^2 + y^2)\{(1-\mu)^2\, x^2 + (1+\mu)^2\, y^2\},$$

i.e.

$$\infty, \quad x = \pm iy, \quad x = \pm\frac{1+\mu}{1-\mu}\, iy,$$

[p. 349] or the curve meets the line ∞ in four points, each of them a point of triple intersection: two of these points are the circular points at ∞.

To find where the circle $x^2 + y^2 = 1$ meets the curve, this gives $x^2 = 1 - y^2$, and thence

$$\begin{aligned} A &= \mu^2\,(\mu^2 - 4) + 4\,(1 + 2\mu^2)\, y^2, \\ B &= 4 - y^2, \\ C &= \mu^2 + 2, \end{aligned}$$

and the equation becomes

$$\begin{aligned} &\{\mu^2\,(\mu^2 - 4) + 4\,(1+2\mu^2)\, y^2\}^3 + 27\mu^4\,(4-y^2)^3\, y^2 - 216\mu^6\,(\mu^2+2)^3\, y^4 \\ &\quad + 54\mu^4\,(\mu^2+2)\, y^2\,(4-y^2)\,\left[\mu^2\,(\mu^2-4) + 4\,(1+2\mu^2)\, y^2\right] = 0, \end{aligned}$$

which is only of the eighth order; it follows that each of the circular points at ∞ (which have been already shown to be points upon the curve) are quadruple points of intersection of the curve and circle. The equation of the eighth order reduces itself to

$$(y^2 - \mu^2)^2\,\{27\mu^4\, y^4 + (\mu^2 - 4)^3\} = 0;$$

the values of x corresponding to the roots $y = \pm\mu$ are obtained without difficulty, and those corresponding to the other roots are at once found by means of the identical equation

$$27\mu^4\, y^4 + (\mu^2 - 4)^3 = -4\,(1+\mu^2)\,(\mu^2-8)^3\, y^2 + \cdots;$$

we thus obtain for the coordinates of the points of intersection of the curve with the circle $x^2 + y^2 = 1$, the values

$$\begin{cases} x = \pm i\mu, \\ y = \pm\mu, \end{cases} \qquad \begin{cases} x = \pm\sqrt{1-\mu^2}\,\dfrac{(1+5\mu^2)}{3\sqrt{3}\,\mu^2}\, i, \\ y = \pm\dfrac{(\mu^2-4)^{\frac{3}{2}}}{3\sqrt{3}\,\mu^2}\, i, \end{cases}$$

each of the points of the first system being a quadruple point of intersection, each of the points of the second system a triple point of intersection, and each of the points of the third system a single point of intersection.

Next, to find where the circle $x^2 + y^2 = \dfrac{1}{\mu^2}$ meets the curve; writing $x^2 = \dfrac{1}{\mu^2} - y^2$, **[p. 350]** we obtain for y an equation of the eighth order, which after all reductions is

$$\left(y^2 - \frac{1}{\mu^2}\right)^2 \left[27\mu^2\, y^4 + (1 - 4\mu^2)^3\right] = 0,$$

and we have for the coordinates of the points of intersection,

$$\begin{cases} x = \pm\dfrac{1}{\mu}\sqrt{1 - \dfrac{1}{\mu^2}}\,, \\ y = \pm\dfrac{1}{\mu^2}\,, \end{cases} \qquad \begin{cases} x = \pm\dfrac{(1+5\mu^2)\sqrt{1-\dfrac{1}{\mu^2}}}{3\sqrt{3}\,\mu^2}\, i, \\ y = \pm\dfrac{(1-4\mu^2)^{\frac{3}{2}}}{3\sqrt{3}\,\mu^2}\, i, \end{cases}$$

each of the points of the first system being a quadruple point of intersection, each of the points of the second system a triple point of intersection, and each of the points of the third system a single point of intersection.

The points of intersection with the axes of x, and the points of triple intersection with the circles $x^2+y^2=1$ and $x^2+y^2=\dfrac{1}{\mu^2}$, are all of them cuspidal points; the two circular points at ∞ are, I think, triple points, and the other two points of intersection with the line ∞, cuspidal points, but I have not verified this: assuming that it is so, there will be a reduction 54 accounted for in the class of the curve, but the curve is, in fact, as will be shown in the sequel, of the class 6; there is consequently a reduction 72 to be accounted for by other singularities of the curve.

XIV.

It is obvious from the preceding formulæ that the caustic stands to the circle, radius $\dfrac{1}{\mu}$, in a relation similar to that in which it stands to the circle, radius 1, i.e. to the refracting circle. In fact, the very same caustic would have been obtained if the circle radius $\dfrac{1}{\mu}$ had been taken for the refracting circle, the index of refraction being $\dfrac{1}{\mu}$ instead of μ. This may be shown very simply by means of the irrational form of the equation as follows.

The equation of the caustic by refraction of the circle, radius 1, index of refraction μ, is as we have seen

$$(1-\mu^2)\,x=\left(1-\mu^{\frac{2}{3}}y^{\frac{2}{3}}\right)^{\frac{3}{2}}+\mu\left(1-\mu^{-\frac{2}{3}}y^{\frac{2}{3}}\right)^{\frac{3}{2}};$$

hence the equation of the caustic by refraction of the circle radius c', index of refraction μ', is

$$(1-\mu'^2)\,\frac{x}{c'}=\left\{1-\mu'^{\frac{2}{3}}\left(\frac{y}{c'}\right)^{\frac{2}{3}}\right\}^{\frac{3}{2}}+\mu'\left\{1-\mu'^{-\frac{2}{3}}\left(\frac{y}{c'}\right)^{\frac{2}{3}}\right\}^{\frac{3}{2}},$$

[p. 351] or, what is the same thing,

$$(1-\mu'^2)\,\frac{x}{c'\mu'}=\frac{1}{\mu'}\left\{1-\mu'^{\frac{2}{3}}c'^{-\frac{2}{3}}y^{\frac{2}{3}}\right\}^{\frac{3}{2}}+\left\{1-\mu'^{-\frac{2}{3}}c'^{-\frac{2}{3}}y^{\frac{2}{3}}\right\}^{\frac{3}{2}},$$

which becomes identical with the equation of the first-mentioned caustic if $\mu'=c'=\dfrac{1}{\mu}$.

Hence taking c instead of 1 as the radius of the first circle, we find,

THEOREM. *The caustic by refraction for parallel rays of a circle, radius c, index of refraction μ, is the same curve as the caustic by refraction for parallel rays of a concentric circle, radius $\dfrac{c}{\mu}$, index of refraction $\dfrac{1}{\mu}$.*

XV.

We may consequently in tracing the caustic confine our attention to the case in which the index of refraction is greater than unity. The circle, radius $\dfrac{c}{\mu}$, will in this case be within the refracting circle, and it is easy to see that if from the extremity of the diameter of the refracting circle perpendicular to the direction of the incident rays, tangents are drawn to the circle, radius $\dfrac{c}{\mu}$, the points of contact are the points of triple intersection of the caustic with the last-mentioned circle, and these points of intersection being, as already observed, cusps, the tangents in question are the tangents to the caustic at these cusps. The points of intersection with the axis of x are also cusps of the caustic, the tangents at these cusps coinciding with the axis of x: two of the last-mentioned cusps, viz. those whose distances from the centre are $\pm\dfrac{1}{\mu+1}$, lie within the circle, radius $\dfrac{c}{\mu}$, the other two of the same four cusps, viz. those whose distances from the centre are $\pm\dfrac{1}{\mu-1}$, lie without the circle, radius $\dfrac{c}{\mu}$; the last-mentioned two cusps lie without the refracting circle, when $\mu<2$, upon this circle, when $\mu=2$, and within it and therefore between the two circles, when $\mu>2$. The caustic is therefore of the forms in the annexed figures 3, 4, 5, in each

[Figure 3: outer (refracting) circle with inner circle of radius c/μ; cusped lens-shaped caustic touching both circles internally; case $\mu < 2$.] [Figure 4: same construction with $\mu = 2$, the outer pair of caustic cusps reaching the refracting circle.] [Figure 5: same with $\mu > 2$, the outer caustic cusps lying between the two circles.]

[p. 352] of which the outer circle is the refracting circle, and μ is > 1, but the three figures correspond respectively to the cases $\mu < 2$, $\mu = 2$ and $\mu > 2$. The same three figures will represent the different forms of the caustic when the inner circle is the refracting circle and $\mu < 1$, the three figures then respectively corresponding to the cases $\mu > \frac{1}{2}$, $\mu = \frac{1}{2}$, and $\mu < \frac{1}{2}$.

XVI.

To find the tangential equation, I retain k instead of its value $\dfrac{1}{\mu}$; the equation of the refracted ray then is

$$x\left(k\cos\phi - \sqrt{1-k^2\sin^2\phi}\right) + y\left(k\sin\phi + \cot\phi\sqrt{1-k^2\sin^2\phi}\right) - k = 0,$$

and representing this by

$$Xx + Yy - k = 0,$$

we have

$$X = k\cos\phi - \sqrt{1-k^2\sin^2\phi},$$
$$Y = k\sin\phi + \cot\phi\sqrt{1-k^2\sin^2\phi},$$

equations which give

$$X\cos\phi + Y\sin\phi = k,$$
$$X^2+Y^2 = \frac{1}{\sin^2\phi},$$

and consequently

$$\sin\phi = \frac{1}{\sqrt{X^2+Y^2}}, \qquad \cos\phi = \frac{\sqrt{X^2+Y^2-1}}{\sqrt{X^2+Y^2}},$$

and we have

$$X\sqrt{X^2+Y^2-1} + Y - k\sqrt{X^2+Y^2} = 0,$$

which gives

$$(X^2+Y^2)(X^2-1-k^2) = -2kY\sqrt{X^2+Y^2};$$

or, dividing out by the factor $\sqrt{X^2+Y^2}$, the equation becomes

$$\sqrt{X^2+Y^2}\,(X^2-1-k^2) = -2kY,$$

from which

$$(X^2+Y^2)(X^2-1-k^2)^2 = 4k^2Y^2;$$

or reducing and arranging, we obtain

$$X^2(X^2-1-k^2)^2 + Y^2(X+1+k)(X+1-k)(X-1+k)(X-1-k) = 0$$

for the tangential equation of the caustic by refraction of a circle for parallel rays. The caustic is therefore of the class 6.

[p. 353]

XVII.

Suppose next that rays proceeding from a point are reflected at a circle.

A very elegant solution of the problem is given by Lagrange in the *Mém. de Turin*; the investigation, as given by Mr P. Smith in a note in the *Cambridge and Dublin Mathematical Journal*, t. II. [1847] p. 237, is as follows:

Let B be the radiant point, RBP an incident ray, and PS a reflected ray; CA a fixed radius; $ACP = \alpha$, $ACB = e$, reciprocal of $CB = c$, reciprocal of $CP = a$. The equations of the incident and reflected ray, where $u = \dfrac{1}{r}$, may be written

$$\begin{aligned} A\sin\theta + B\cos\theta &= u; \quad \text{incident ray,} \\ A\sin(2\alpha-\theta) + B\cos(2\alpha-\theta) &= u; \quad \text{reflected ray,} \end{aligned}$$

the conditions for determining A and B being

$$\begin{aligned} a &= A\sin\alpha + B\cos\alpha, \\ c &= A\sin e + B\cos e, \end{aligned}$$

whence

$$A = \frac{a\cos e - c\cos\alpha}{\sin(\alpha - e)}, \qquad B = \frac{c\sin\alpha - a\sin e}{\sin(\alpha - e)}.$$

[Figure: circle with centre C; radiant point B inside; incident ray RBP meeting the circle at P and reflected as PS; fixed radius CA.]

Substituting these values, the equation of the reflected ray becomes

$$a\sin(2\alpha - \theta - e) = u\sin(\alpha - e) + c\sin(\alpha - \theta),$$

from which and its differential with respect to the arbitrary parameter α, the equation of the caustic, or envelope of the reflected rays, will be found by eliminating α.

In this, α being the only quantity treated as variable in the differentiation, let

$$2\alpha - \theta - e = 2\phi,$$

therefore

$$\alpha = \phi + \tfrac{1}{2}(\theta + e),$$

[p. 354] and the equation becomes

$$a\sin 2\phi = u\sin\{\phi + \tfrac{1}{2}(\theta - e)\} + c\sin\{\phi - \tfrac{1}{2}(\theta - e)\}.$$

Make

$$P = \frac{(u+c)\cos\frac{1}{2}(\theta - e)}{2a}, \qquad Q = \frac{(u-c)\sin\frac{1}{2}(\theta - e)}{2a},$$

also

$$x = \frac{1}{\cos\phi}, \qquad y = \frac{1}{\sin\phi},$$

then the equation becomes

$$Px + Qy = 1,$$

with the condition

$$x^{-2} + y^{-2} = 1.$$

Hence

$$P = \lambda x^{-3}, \qquad Q = \lambda y^{-3};$$

multiplying by x and y, and adding, we find $\lambda = 1$; therefore

$$P = x^{-3}, \qquad Q = y^{-3}.$$

Hence

$$P^{\frac{2}{3}} + Q^{\frac{2}{3}} = 1,$$

or restoring the values of P and Q,

$$\big\{(u+c)\cos\tfrac{1}{2}(\theta-e)\big\}^{\frac{2}{3}}+\big\{(u-c)\sin\tfrac{1}{2}(\theta-e)\big\}^{\frac{2}{3}}=1,$$

the equation of the caustic.

XVIII.

But the equation of the caustic for rays proceeding from a point and reflected at a circle may be obtained by a different method, as follows:

Take the centre of the circle for origin; let c be the radius of the circle, a, b the coordinates of the radiant point, α, β the coordinates of the point of incidence, x, y the coordinates of a point in the reflected ray. Then we have from the equation of the circle $\alpha^2+\beta^2=c^2$, and the equation of the reflected ray is by the general formula,

$$(b\alpha-a\beta)(\alpha x+\beta y-c^2)+(y\alpha-x\beta)(a\alpha+b\beta-c^2)=0;$$

[p. 355] or arranging the terms in a different order,

$$(bx+ay)(\alpha^2-\beta^2)+2(by-ax)\,\alpha\beta-c^2(b+y)\,\alpha+c^2(a+x)\,\beta=0;$$

and writing herein $\alpha=c\cos\theta$, $\beta=c\sin\theta$, the equation becomes

$$(bx+ay)\cos 2\theta+(by-ax)\sin 2\theta-(b+y)\,c\cos\theta+(a+x)\,c\sin\theta=0,$$

where θ is a variable parameter.

Now in general to find the envelope of

$$A\cos 2\theta+B\sin 2\theta+C\cos\theta+D\sin\theta+E=0,$$

we may put $z=e^{i\theta}$, which gives the equation

$$(A-Bi)\,z^4+(C-Di)\,z^3+2E\,z^2+(C+Di)\,z+(A+Bi)=0,$$

and equate the discriminant to zero: this gives

$$(4I)^3-27\,(-8J)^2=0,$$

where

$$\begin{aligned}4I&=4\,(A^2+B^2)-(C^2+D^2)+4E^2,\\ -8J&=A\,(C^2-D^2)+2BCD-\big[8\,(A^2+B^2)+(C^2+D^2)\big]\,E+4E^3,\end{aligned}$$

and consequently

$$\begin{aligned}&\big[4\,(A^2+B^2)-(C^2+D^2)+4E^2\big]^3\\ &\quad-\,27\big\{A\,(C^2-D^2)+2BCD-\big[8(A^2+B^2)+(C^2+D^2)\big]\,E+4E^3\big\}^2=0;\end{aligned}$$

and substituting for A, B, C, D, E their values, we find

$$\big\{4\,(a^2+b^2)(x^2+y^2)-c^2\big[(x+a)^2+(y+b)^2\big]\big\}^3-27\,(bx-ay)^2\,(x^2+y^2-a^2-b^2)^2=0,$$

for the equation of the caustic in the case of rays proceeding from a point and reflected at a circle: the equation was first obtained by St Laurent.

It will be convenient to consider the axis of x as passing through the radiant point; this gives $b=0$; and if we assume also $c=1$, the equation of the caustic becomes

$$\big\{(4a^2-1)(x^2+y^2)-2ax-a^2\big\}^3-27a^2y^2\,(x^2+y^2-a^2)^2=0.$$

XIX.

Reverting to the equation of the reflected ray, and putting, as before, $c = 1$, $b = 0$, this becomes

$$(-2a\cos\theta + 1)\, x + \frac{a\cos 2\theta - \cos\theta}{\sin\theta}\, y + a = 0;$$

differentiating with respect to θ, we have

$$2a\sin\theta \cdot x + \frac{-2a\sin\theta\sin\theta\,(1 - 2\sin^2\theta) + (a\cos 2\theta - \cos\theta)\cos\theta}{\sin^2\theta}\, y = 0;$$

[p. 356] and from these equations

$$x = \frac{a^2\cos\theta\,(1 + 2\sin^2\theta) - a}{1 - 3a\cos 2\theta + 2a^2},$$
$$y = \frac{2a^2\sin^3\theta}{1 - 3a\cos 2\theta + 2a^2},$$

which give the coordinates of a point of the caustic in terms of the angle θ which determines the position of the point of incidence. The values in question satisfy, as they should do, the equation

$$\left\{(4a^2 - 1)(x^2 + y^2) - 2ax - a^2\right\}^3 - 27a^2y^2\,(x^2 + y^2 - a^2)^2 = 0;$$

we have, in fact,

$$x^2 + y^2 - a^2 = \frac{-4a^2\,(\cos\theta - a)^2}{(1 - 3a\cos 2\theta + 2a^2)^2},$$
$$(4a^2 - 1)(x^2 + y^2) - 2ax - a^2 = \frac{12a^2\,(\cos\theta - a)^2}{(1 - 3a\cos 2\theta + 2a^2)^2},$$

from which it is easy to derive the equation in question.

XX.

If we represent the equation of the reflected ray by

$$Xx + Yy + a = 0,$$

then we have

$$X = -2a\cos\theta + 1,$$
$$Y = \frac{a\cos 2\theta - \cos\theta}{\sin\theta},$$

and thence

$$(X - 1)^2 - 4a^2 = -4a^2\sin^2\theta,$$
$$X^2 + Y^2 = \frac{1}{\sin^2\theta}\,(1 - 2a\cos\theta + a^2),$$
$$X + a^2 = 1 - 2a\cos\theta + a^2,$$

and consequently

$$(X^2 + Y^2)\left\{(X - 1)^2 - 4a^2\right\} + 4a^2X + 4a^4 = 0,$$

or, what is the same thing,

$$\left\{X(X - 1) - 2a^2\right\}^2 + Y^2\left\{(X - 1)^2 - 4a^2\right\} = 0,$$

which may be considered as the tangential equation of the caustic by reflexion of a circle; or if we consider X, Y as the coordinates of a point, then the equation may be considered as that of the polar of the caustic.

The polar is therefore a curve of the fourth order, having two double points defined by the equations $X(X-1)-2a^2=0$, $Y=0$, **[p. 357]** and a third double point at infinity on the axis of Y, i.e. three double points in all; the number of cusps is therefore 0, and there are consequently 4 double tangents and 6 inflections, and the curve is of the class 6. And as Y is given as an explicit function of X, there is of course no difficulty in tracing the curve. We thus see that the caustic by reflexion of a circle is a curve of the order 6, and has 4 double points and 6 cusps (the circular points at infinity are each of them a cusp, so that the number of cusps at a finite distance is 4): this coincides with the conclusions which will be presently obtained by considering the equation of the caustic.

XXI.

The equation of the caustic by reflexion of a circle is

$$\left\{(4a^2-1)(x^2+y^2)-2ax-a^2\right\}^3-27a^2y^2\,(x^2+y^2-a^2)^2=0.$$

Suppose first that $y=0$, we have

$$\left\{(4a^2-1)\,x^2-2ax-a^2\right\}^3=0,$$

$$x=\frac{a}{2a-1},\qquad x=\frac{-a}{2a+1},$$

or the curve meets the axis of x in two points, each of which is a triple point of intersection.

Write next $x^2=y^2+a^2$, this gives

$$\left\{(4a^2-1)\,a^2-2ax-a^2\right\}^3=0,$$

and consequently

$$x=-a\,(1-2a^2),\qquad y=\pm\,2a^2\sqrt{1-a^2},$$

or the curve meets the circle $x^2+y^2-a^2=0$ in two points, each of which is a triple point of intersection.

To find the nature of the infinite branches, we may write, retaining only the terms of the degrees six and five,

$$(x^2+y^2)^2\left[(4a^2-1)^3(x^2+y^2)\right]-6\,(4a^2-1)^2\,a\,(x^2+y^2)^2\,x-27a^2y^2\,(x^2+y^2)^2=0;$$

and rejecting the factor $(x^2+y^2)^2$, this gives

$$(4a^2-1)^3\,x^4-\left\{(4a^2-1)^3-27a^2\right\}y^4-6\,(4a^2-1)^2\,a\,x^3=0;$$

or reducing,

$$(4a^2-1)^3\,x^4-(1-a^2)\,(8a^2-1)^3\,y^4-6\,(4a^2-1)^2\,a\,x^3=0;$$

and it follows that there are two asymptotes, the equations of which are

$$y=\pm\frac{(4a^2-1)^{\frac{3}{4}}}{(1-a^2)^{\frac{1}{4}}\,(8a^2+1)^{\frac{3}{4}}}\left(x-\frac{3a}{4a^2-1}\right).$$

[p. 358] Represent for a moment the equation of one of the asymptotes by $y=A\,(x-\alpha)$, then the perpendicular from the origin or centre of the reflecting circle is $\dfrac{A\alpha}{\sqrt{1+A^2}}$, and

$$A\alpha=\frac{3a\sqrt{4a^2-1}}{\sqrt{1-a^2}\,(1+8a^2)^{\frac{1}{2}}},$$

$$1+A^2=\frac{(1-a^2)(1+8a^2)^2+(4a^2-1)^3}{(1-a^2)(1+8a^2)^2}=\frac{27a^2}{(1-a^2)(1+8a^2)^2},$$

$$\sqrt{1+A^2}=\frac{3\sqrt{3}\,a}{\sqrt{1-a^2}\,(1+8a^2)^{\frac{1}{2}}},$$

and the perpendicular is $\frac{1}{\sqrt{3}}\sqrt{4a^2-1}$, which is less than a if only $a^2 < 1$, i.e. in every case in which the asymptote is real.

The tangents parallel and perpendicular to the axis of x are most readily obtained from the equation of the reflected ray, viz.

$$(-2a\cos\theta + 1)\,x + \frac{a\cos 2\theta - \cos\theta}{\sin\theta}\,y + a = 0;$$

the coefficient of x (if the equation is first multiplied by $\sin\theta$) vanishes if $\sin\theta = 0$, which gives the axis of x, or if $\cos\theta = \dfrac{1}{2a}$, which gives $y = \pm\dfrac{\sqrt{4a^2-1}}{2a}$, for the tangents parallel to the axis of x.

The coefficient of y vanishes if $a\cos 2\theta - \cos\theta = 0$; this gives

$$\cos\theta = \frac{1 \pm \sqrt{8a^2+1}}{4a}, \qquad \sin\theta = \frac{1}{8a^2}\left(4a^2 - 1 \mp \sqrt{8a^2+1}\right),$$

and the tangents perpendicular to the axis of x are thus given by

$$x = \frac{-2a}{1+\sqrt{8a^2+1}};$$

these tangents are in fact double tangents of the caustic. In order that the point of contact may be real, it is necessary that $\sin\theta$, $\cos\theta$ should be real; this will be the case for both values of the ambiguous sign if $a >$ or $= 1$, but only for the upper value if $a < 1$.

It has just been shown that for the tangents parallel to the axis of x, we have

$$y = \pm\frac{\sqrt{4a^2-1}}{2a},$$

the values of y being real for $a > \frac{1}{2}$: it may be noticed that the value $y = \dfrac{\sqrt{4a^2-1}}{2a}$ **[p. 359]** is greater, equal, or less than or to $y = 2a^2\sqrt{1-a^2}$, according as $a >, =,$ or $< \dfrac{1}{\sqrt{2}}$; this depends on the identity $(4a^2-1) - 16a^4\,(1-a^2) = (2a^2-1)^2\,(7a^2+1)$.

To find the points of intersection with the reflecting circle, $x^2+y^2-1=0$, we have

$$(3a^2-1-2ax)^3 - 27a^2\,(1-a^2)\,(1-x^2)^2 = 0;$$

or, reducing,

$$8a^3x^3 + (-27a^4+18a^2-15)\,a^2x^2 + (54a^4-36a^2+6)\,a\,x + (-27a^4+18a^2+1) = 0,$$

i.e.

$$(ax-1)^2\,(8ax - 27a^4 + 18a^2 + 1) = 0.$$

The factor $(ax-1)^2$ equated to zero shows that the caustic touches the circle in the points $x = \dfrac{1}{a}$, $y = \pm\sqrt{1-\dfrac{1}{a^2}}$, i.e. in the points in which the circle is met by the polar of the radiant point, and which are real or imaginary according as $a >$ or < 1. The other factor gives

$$x = \frac{27a^4 - 18a^2 - 1}{8a}.$$

Putting this value equal to ± 1, the resulting equation is $(a+1)\,(27a^3+9a+1) = 0$, and it follows that x will be in absolute magnitude greater or less than 1, i.e. the points in question will be imaginary or real, according as $a > 1$ or $a < 1$.

It is easy to see that the curve passes through the circular points at infinity, and that these points are cusps on the curve; the two points of intersection with the axis of x are cusps (the axis of x being the tangent), and the two points of intersection with the circle $x^2+y^2-a^2=0$ are also cusps, the tangent at each of the cusps coinciding with the tangent of the circle; there are consequently in all 6 cusps.

XXII.

To investigate the position of the double points we may proceed as follows: write for shortness $P = (4a^2 - 1)(x^2 + y^2) - 2ax - a^2$, $Q = ay$, $S = x^2 + y^2 - a^2$; the equation of the caustic is

$$P^3 - 27\,Q^2\,S^2 = 0;$$

hence, at a double point,

$$P^2\,\frac{dP}{dx} - 18\,Q\,S^2\,\frac{dQ}{dx} = 0, \qquad P^2\,\frac{dP}{dy} - 18\,Q\,S^2\,\frac{dQ}{dy} = 0,$$

one of which equations may be replaced by

$$\frac{dP}{dx}\,\frac{dQ}{dy} - \frac{dP}{dy}\,\frac{dQ}{dx} = 0.$$